

INDIAN INSTITUTE OF TECHNOLOGY, BOMBAY

Department of Mathematics

SI 427 (Probability Theory)

Tutorial Sheet-I

1. Let $\Omega \neq \emptyset$ be a finite set and $\mathcal{F}$ is a field of subsets of Ω . Show that $\mathcal{F}$ is a σ -field.
2. Write down the expression in set notation corresponding to each of the following events.
 - (i) The event which occurs if exactly one of the events A, B occurs.
 - (ii) The event which occurs if none of the events A, B, C occurs.
 - (iii) The event which occurs if exactly two of the events A, B, C occurs.

3. Let $(\Omega, \mathcal{F}, P)$ be a probability space and A, B, C be events in $\mathcal{F}$ such that

$$P(A) = 0.7, P(B) = 0.6, P(C) = 0.5, P(AB) = 0.4,$$

$$P(AC) = 0.3, P(BC) = 0.2 \text{ and } P(ABC) = 0.1.$$

Find (i) $P(A \cup B \cup C)$ (ii) $P(A^c C)$ (iii) $P(A^c B^c C^c)$.

4. Does there exists a probability measure P such that the events A, B, C satisfies

$$P(A) = 0.6, P(B) = 0.8, P(C) = 0.7, P(AB) = 0.5,$$

$$P(AC) = 0.4, P(BC) = 0.5 \text{ and } P(ABC) = 0.1?$$

5. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $A_1, A_2 \dots A_n$ be events in $\mathcal{F}$. Show that

$$P(A_1 \dots A_n) \geq \sum_{i=1}^n P(A_i) - n + 1.$$

6. Define $P : \mathcal{P}(\mathbb{N}) \rightarrow [0, 1]$ as follows.

$$P(A) = \begin{cases} 0 & \text{if } A \text{ is finite} \\ 1 & \text{if } A \text{ is infinite} \end{cases}$$

Is P a probability measure? Justify your answer.

7. Let Ω be a nonempty set and $\mathcal{F}$ is a σ -field of subsets of Ω . Let $P : \mathcal{F} \rightarrow [0, 1]$ be such that

- (i) $P(\Omega) = 1$
- (ii) For $A_1, A_2 \in \mathcal{F}$ disjoint

$$P(A_1 \cup A_2) = P(A_1) + P(A_2)$$

- (iii) Whenever $\{A_n\}$ is a decreasing sequence of events from $\mathcal{F}$, $\lim_{n \rightarrow \infty} P(A_n) = P(\cap_{i=1}^{\infty} A_i)$.

Then show that P is a probability measure.

8. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $A_1, A_2, \dots \in \mathcal{F}$ be such that $P(A_i A_j) = 0$, $i \neq j$. Show that

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

9. Let Ω be an uncountable set and

$$\mathcal{F} = \{A \subseteq \Omega \mid \text{either } A \text{ is countable or } A^c \text{ is countable}\}.$$

Show that $\mathcal{F}$ is a σ -field.

10. Let $\mathcal{D}_1, \mathcal{D}_2$ be two nonempty family of subsets of a nonempty set Ω such that $\mathcal{D}_1 \subset \mathcal{D}_2$. Show that $\sigma(\mathcal{D}_1) \subseteq \sigma(\mathcal{D}_2)$ Is the inclusion always strict?
11. In a group of n ‘unrelated’ individuals, none born on a leap year, show that the probability that atleast two share a birth day exceeds half if $n \geq 23$.

12. (Maxwell-Boltzmann Statistics) A configuration of n balls placed in r urns is an arrangement $(k_1, \dots, k_r)$ where k_i denotes the number of balls in the i th urn and $k_1 + \dots + k_r = n$. If the balls and urns are distinguishable and the balls are distributed at random into the urns, find the probability of a particular configuration.
13. Let $\pi = (\pi_1, \pi_2, \dots, \pi_{52})$ denote a permutation of numbers from 1 to 52 done at random (You may think it as a perfectly shuffled deck of cards). By coincidence at a location k , we mean $\pi_k = k$. What is the probability that there is atleast one coincidence.
14. Does there exists a σ -field with number of elements 4098? Justify your answer.